

Kuanwei Chen

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Education

National Central University, Taoyuan, Taiwan

Starting Sep 2025

Master in Computer Science, specialization in Artificial Intelligence

National Changhua University of Education, Changhua, Taiwan

Sep 2021 – Jun 2025

Bachelor in Electrical Engineering, GPA 3.8/4.0 | Grad. Cert.

Experience

Project Assistant in AI Application Laboratory

Jun 2024 – Dec 2024

Certificate

- Assisted in TEEP and IIPP projects, collaborating with 8 international interns.
- Contributed to “Multi-biometric Authentication Systems” and “Action Recognition Systems”.

Research Publication

STRLite: Lightweight Masked Autoencoders for Adaptable Scene Text Recognition

IEEE ICCE-TW

ResearchGate | GitHub

- Co-first author; designed a 6M-parameter Vision Transformer (ViT) with Masked Autoencoder (MAE) pretraining and fine-tuning with an autoregressive decoder for Scene Text Recognition.
- Achieved 93.82% weighted accuracy on six standard Computer Vision benchmarks and 81.03% on U14M.

Towards Compact Sign Language Translation: Frame Rate and Model Size Trade-offs

IEEE ICCE-TW

arXiv | ResearchGate

- First author; built a 77M-parameter end-to-end Natural Language Processing (NLP) pipeline.
- reduce computation by ~75% with minimal BLEU impact (10.06→9.53), providing efficiency trade-off insights.

Two-step Authentication: Multi-biometric System Using Voice and Facial Recognition

IET ICETA

Publication | arXiv | ResearchGate | GitHub | Certificate

- First author; led a team of 2 interns in designing a multi-modal AI/ML system combining Machine Vision and Audio Recognition/Speaker Verification.
- Designed a ResNet-based Convolutional Neural Network (CNN) with Fbank feature extraction and triplet loss for voice verification (99.1%); fine-tuned VGG16 with MTCNN object detection in TensorFlow (95.1%).

Projects

SignDATA: Data Pipeline for Sign Language Translation

Python, Det, Pose, System Design

arXiv | ResearchGate | GitHub

- First author; open-source repo with 100+ stars; built a config-driven data preprocessing, and dataset management system for standardizing heterogeneous sign-language corpora for AI/ML Pipeline development.
- Designed processing pipeline integrating YOLO/MMDet detection and MediaPipe/MMPose landmark extraction for structured feature engineering, applying system-design principles for multi-dataset scalability.

Skills

Languages

English (IELTS 6.5, TOEIC 830), Mandarin (native), Hokkien (native)

Programming

Python, C/C++, HTML, CSS, JavaScript

ML/DL Libraries

PyTorch, TensorFlow, Hugging Face, Scikit-learn, NumPy, Pandas

Frontend & Tools

React, Vue, Git, GitHub

Research Areas

Computer Vision, Natural Language Processing (NLP), Deep Learning, Data Cleaning, Data Analysis

Awards and Certifications

Micro-Program in Green Energy Technology

Electrical Engineering, NCUE

- Courses included “Introductions to Green Energy Technology,” “Circuit Theory Experiments,” “Circuit Theory,” and “Special Topic for Industry.”
- Gained foundational knowledge in Renewable Energy Systems, Power Electronics, and Circuit Analysis for sustainable engineering applications.

Teaching Assistant – Linear Algebra [Certificate]

Electrical Engineering, NCUE

- Created and delivered supplementary materials for 40+ students and assisted with exam invigilation.

Model United Nations Certifications

2018–2019

HSNU MUN VII | TCGSMUN VII | Central Taiwan MUN IX | Central Taiwan MUN XII